



Midterm Examination

الاختبار النصفى

Course Code / رمز المقرر	ITDB 211	Course / المقرر	Database Systems نظم قواعد البيانات
Semester / الفصل	First Semester	Academic Year / العام	1448 AH / 2026
Duration / الزمن	75 minutes	Total Marks / الدرجة	30
Sections / الشعب	All Sections		

Student Name / اسم الطالب:		University ID / الرقم الجامعي:	
-------------------------------	--	-----------------------------------	--

Instructions / التعليمات

- Answer all questions in the spaces provided. أجب عن جميع الأسئلة.
- Write clearly and show calculation steps when required.
- Mobile phones and unauthorized materials are not permitted.
- Marks shown beside a question are the official marks for that item.

هذا مستند اصطناعي لاختبار النظام فقط، وليس صادرًا عن جامعة حقيقية أو جهة اعتماد رسمية.

Question 1 - Multiple Choice (8 marks)

Q1.1 A relation is in Second Normal Form when it is in 1NF and has no:

[1 marks]

- Transitive dependency
- Partial dependency
- Foreign key
- Composite attribute

Q1.2 Which SQL clause filters groups after aggregation?

[1 marks]

- WHERE
- GROUP BY
- HAVING
- ORDER BY

Q1.3 A minimal superkey is called a:

[1 marks]

- A. Candidate key
- B. Secondary key
- C. Surrogate key
- D. Foreign key

Q1.4 Which operation combines rows from two relations based on a condition?

[1 marks]

- A. Projection
- B. Selection
- C. Join
- D. Division

Q1.5 The ACID property that prevents partial transactions is:

[1 marks]

- A. Consistency
- B. Isolation
- C. Atomicity
- D. Durability

Q1.6 Which command removes a table definition and its data?

[1 marks]

- A. DELETE
- B. TRUNCATE
- C. DROP
- D. REMOVE

Q1.7 A foreign key mainly enforces:

[1 marks]

- A. Entity integrity
- B. Referential integrity
- C. Domain independence
- D. Physical independence

Q1.8 Which index structure is commonly suitable for range queries?

[1 marks]

- A. Hash index
- B. B+ tree
- C. Heap file
- D. Bitmap only

Question 2 - True or False (5 marks)

No.	Statement	T / F
1	Every candidate key must be selected as the primary key.	
2	A foreign key may contain NULL when the relationship is optional.	
3	Projection chooses rows from a relation.	
4	Third Normal Form removes transitive dependency on a key.	
5	COMMIT makes the current transaction permanent.	

Question 3 - Short Answer (8 marks)

Q3(a) Distinguish between a candidate key and a primary key.

[3 marks]

.....

.....

Q3(b) Explain partial dependency and give one relational example.

[3 marks]

.....

.....

Q3(c) State two practical benefits of database normalization.

[2 marks]

.....

Question 4 - SQL Application (9 marks)

Use the following schema:

```
STUDENT(StudentID, StudentName, Major)
COURSE(CourseID, CourseTitle, Credits)
ENROLLMENT(StudentID, CourseID, Semester, Grade)
```

Q4(a) Write an SQL query to list the names of IT students.

[3 marks]

.....

.....

Q4(b) Write an SQL query to display each student name with the titles of courses in which the student is enrolled.

[3 marks]

.....

.....

Q4(c) Write an SQL query to show the number of students enrolled in each course.

[3 marks]

.....

.....